

Cybersecurity Prevention

Lesson 44 — Controls and the CIA Triad

JKSSB Computer Awareness

October 8, 2026

Why Prevention Comes First

- ▶ Removing a virus after infection is costly; **blocking it first** is cheaper and safer.
- ▶ Prevention uses small daily habits plus a few tools that each do **one distinct job**.
- ▶ The exam favourite is matching: **which control does what**.
- ▶ Learn each control as name → threat it stops → what it cannot do.

The one idea

No single tool stops everything; layered controls protect the CIA triad.

The CIA Triad

- ▶ **Confidentiality**: only authorised persons can read the data.
- ▶ **Integrity**: the data is accurate and has not been changed by unauthorised means.
- ▶ **Availability**: the data and systems are reachable when needed.

One-line memory hook

Confidentiality keeps secrets secret; integrity keeps data correct; availability keeps systems up.

CIA Triad: Which Attack Breaks What

Goal broken	Example attack	Protective control
Confidentiality	Stealing passwords over open Wi-Fi	Encryption, strong passwords, MFA
Integrity	Altering marks in a file	Clean backups, digital signatures, updates
Availability	Ransomware locks files	Backup, antivirus, firewall

One attack can break more than one goal at once.

Antivirus: Signatures

- ▶ **Antivirus** software detects, blocks, and removes malicious programs (malware).
- ▶ A **signature** is a stored pattern of a known virus, like a fingerprint on record.
- ▶ The antivirus compares each file against its **signature database** of known threats.
- ▶ New viruses appear daily, so signatures must be **updated regularly**; an outdated database misses new malware.

Trap alert

Signatures catch **known** malware. Unknown (zero-day) malware needs behaviour-based detection instead.

Real-Time Scanning and Quarantine

- ▶ **Real-time scanning** checks files as they are opened, downloaded, or copied — before harm is done.
- ▶ A scheduled **full scan** checks the whole system on demand.
- ▶ **Quarantine** isolates a suspicious file in a safe separate area so it cannot run or spread.
- ▶ Quarantine is **not** deletion: the file is held safely until the user deletes or restores it.

Keep the pair straight

Quarantine isolates; deletion removes. Quarantined files can be restored; deleted files cannot.

Firewall: The Gatekeeper

- ▶ A **firewall** monitors incoming and outgoing network traffic and blocks what its rules forbid.
- ▶ It stops **unauthorised network access**, such as a stranger reaching into the office computer.
- ▶ A firewall can be **software** (on the computer) or **hardware** (on the network boundary).
- ▶ It does **not** remove viruses already on the disk — that is the antivirus job.

Trap alert

Firewall blocks bad **traffic**; antivirus removes bad **files**. Do not swap them.

Patching and Updates

- ▶ A **patch** is a small fix released by the vendor for a known security weakness.
- ▶ **Software updates** close the holes that attackers exploit, so install them promptly.
- ▶ Update the **operating system, browser, and antivirus** alike; all three have exploitable flaws.
- ▶ Postponing updates leaves known doors open.

Office desktop example

The office PC shows an operating-system update notice. Installing it patches a known weakness; clicking “remind me later” for weeks leaves the machine exposed.

Strong Passwords

- ▶ A **strong password** is long and mixes uppercase, lowercase, numbers, and symbols.
- ▶ Never reuse one password across sites, and never share it.
- ▶ Change **default passwords** on routers and devices at once.
- ▶ A **password manager** stores distinct passwords so the user remembers only one master secret.

Weak vs strong

rahul123 is weak; RaHuL#2026!Desk is strong: longer, mixed, and unpredictable.

MFA: A Second Proof

- ▶ **MFA** stands for Multi-Factor Authentication.
- ▶ It asks for **two or more proofs**: something you know (password) plus something you have (phone code) or are (fingerprint).
- ▶ Even if the password leaks, the attacker still lacks the **second factor**.
- ▶ One-time passwords (OTPs) sent to the phone are the most common second factor.

Trap alert

A password alone is one factor. MFA is the **extra** check, not a stronger password.

Backup: The Availability Control

- ▶ A **backup** is a separate clean copy of important data, used when the original is lost, damaged, or locked by ransomware.
- ▶ It protects **availability**: work continues even after data loss.
- ▶ Keep backups **regular** and **offline or off-site**, so one incident cannot destroy both copies.
- ▶ Test restores: a backup that cannot be restored is **not** a backup.

Keep the pair straight

Backup restores **lost** data; antivirus removes **malicious** data.

Safe Browsing

- ▶ Look for **HTTPS** and the padlock before entering passwords or bank details.
- ▶ Do not click unknown links or attachments; **phishing** messages imitate trusted senders.
- ▶ Download software only from **official or trusted sources**.
- ▶ Log out of shared computers and avoid saving passwords on them.

Phishing example

A message claims the bank blocked the account and demands an instant login through its link. A safe user ignores the link and opens the bank site by typing the address directly.

Secure Wi-Fi

- ▶ Home and office Wi-Fi must use **WPA2 or WPA3** encryption with a strong password.
- ▶ **Open (password-free) public Wi-Fi** exposes traffic; attackers can read what passes over it.
- ▶ Avoid banking or official logins on open hotspots; use mobile data or a trusted **VPN** instead.
- ▶ Turn off auto-connect and change the router's **default admin password**.

Trap alert

The padlock (HTTPS) protects the website session; WPA2/WPA3 protects the **wireless link**. Both layers matter.

Removable-Media Safety

- ▶ Scan every **USB drive or memory card** with antivirus **before** opening its files.
- ▶ Disable **autorun** so a plugged-in drive cannot launch programs by itself.
- ▶ Do not plug unknown found drives into the office computer.
- ▶ **Eject** the device safely before removal to avoid corrupting files.

Office desktop example

A visitor offers files on a pen drive. The safe steps are: scan first, open only the needed files, then eject — never let it auto-run.

Which Control Does What

Control	Main job
Antivirus + signatures	Detect and remove known malware files
Firewall	Block unauthorised network traffic
Patching / updates	Close known software weaknesses
Strong password + MFA	Prove identity; block account theft
Backup	Restore data after loss or ransomware
HTTPS + safe browsing	Protect sessions; dodge phishing and bad downloads
WPA2/WPA3 + VPN	Protect data on the network link
Scan USB + disable autorun	Stop malware entering on drives

Read the stem's verb: remove, block, prove, restore, or protect.

Trap Alert: Controls to Keep Straight

- ▶ Firewall blocks **traffic**; antivirus removes **files**.
- ▶ Quarantine **isolates**; deletion **removes**.
- ▶ MFA adds a **second proof**; a long password is still **one factor**.
- ▶ Backup restores **availability**; encryption protects **confidentiality**.
- ▶ HTTPS protects the **session**; WPA2/WPA3 protects the **Wi-Fi link**.

Lesson 44: Recall

- ▶ CIA triad: confidentiality (secrets kept), integrity (data correct), availability (systems reachable).
- ▶ Antivirus uses signatures for known malware; real-time scanning checks on access; quarantine isolates.
- ▶ Firewall blocks unauthorised traffic; it does not clean infected files.
- ▶ Patch promptly; use long unique passwords plus MFA (Multi-Factor Authentication).
- ▶ Keep regular offline backups; browse over HTTPS, avoid open Wi-Fi for sensitive logins, and scan USB drives with autorun off.